

Question	Answer	Marks
2(a)	energy (of a mass/body/object) due to motion/speed/velocity	B1
2(b)(i)	$E = \frac{1}{2}mv^2$	C1
	$480 = \frac{1}{2} \times m \times 80^2$ so $m = 0.15 \text{ kg}$	A1
2(b)(ii)	1. $E = mgh$ or $\Delta E = mg\Delta h$	C1
	$= 0.15 \times 9.81 \times 210$ $= 310 \text{ J}$	A1
	2. work done $= 480 - 310$ $= 170 \text{ J}$	A1
2(b)(iii)	work done $= Fs$	C1
	force $= 170/210$ $= 0.81 \text{ N}$	A1
2(b)(iv)	curved line from positive value on v -axis to $(T, 0)$	M1
	magnitude of gradient decreases	A1
2(b)(v)	as shell rises force decreases and as shell falls force increases	B1
	as shell rises force is downward and as shell falls force is upward	B1
	or	
	as shell rises the force decreases and is downward	(B1)
	as shell falls the force increases and is upward	(B1)

9702/23

Cambridge International AS/A Level – Mark Scheme

October/November 2018

PUBLISHED

Question	Answer	Marks
2(c)	(resultant) force proportional/equal to rate of change of momentum	B1